Benchmark: **MA.7.AR.3.2** · Solve problems involving proportions Student code: _____ Calculator: yes

Learning goal: I can set up and solve proportions for real-world problems. (circle your level after scoring)

4	I can solve a multi-step problem with proportions and explain why the setup makes sense.
3	I can write a proportion from a situation and solve it.
2	I can tell whether two ratios are equivalent.
1	I need help to start this benchmark.

1. In a trail mix, the ratio of peanuts to raisins is $28{:}16$. A hiker uses 20 ounces of raisins.

Write a proportion that can be used to find p, the number of ounces of peanuts that keeps the same ratio. Then solve the proportion.

2. A drip line uses 3.6 gallons of water every 4 hours.

How many gallons of water does the drip line use in 9 hours?

Turn over →

Benchmark: **MA.7.AR.3.2** · Solve problems involving proportions Student code: _____

3. The table shows what different amounts of ground beef cost at one store. The cost is proportional to the number of pounds.

Pounds of ground beef	Cost
2	$8.88
2.5	$11.10
3	$13.32
4.5	$19.98

At this store, what will 6 pounds of ground beef cost?

4. A recipe uses $\frac{1}{2}$ cup of oil for every 2 dozen muffins. The amounts are proportional.

How much oil is needed for 5 dozen muffins? Write your answer as a fraction or a mixed number in simplest form.

5. Tomás usually runs 10-kilometer races, and 10 kilometers is about 6.2 miles. Tomás is training for a trail race, which is 15 miles. Tomás runs at a steady pace of 6 minutes per kilometer.

About how many kilometers is the trail race, to the nearest tenth? About how long will Tomás take to run it at that pace, to the nearest minute?

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-AR.3.2 · MA.7.AR.3.2 · Solve problems involving proportions

Unit 6: Multi-Step Problems With Proportional Relationships. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular $\frac{28}{16} = 1\frac{12}{16} = \frac{p}{20}$ **(or an equivalent proportion); $p = 35$. Score: 1 point for the proportion (or an equivalent proportion) and its solution; the solution alone earns 0.**

Keep peanuts over raisins on both sides: $\frac{28}{16} = \frac{p}{20}$.

$28{:}16$ in simplest form is $7{:}4$, and $20 \div 4 = 5$, so both parts are multiplied by 5: $p = 7 \times 5 = 35$.

Check by cross-multiplying: $28 \times 20 = 560$ and $16 \times 35 = 560$.

2. Regular **8.1 gallons**

Unit rate: $3.6 \div 4 = 0.9$ gallons per hour.

$9 \times 0.9 = 8.1$.

3. Regular **$26.64**

The cost is proportional to the weight, so find the cost of one pound.

$\$8.88 \div 2 = \$4.44$ for one pound.

$\$4.44 \times 6 = \$26.64$.

4. Regular $1\frac{1}{4} = \frac{5}{4}$ **cups**

Oil for 1 dozen: $\frac{1}{2} \div 2 = \frac{1}{4}$ cup.

Oil for 5 dozen: $5 \times \frac{1}{4} = 1\frac{1}{4}$ cups.

5. Rigor **about 24.2 kilometers; about 145 minutes (2 hours 25 minutes). Score: 1 point for the distance and the time; one alone earns 0.**

Kilometers: set up a proportion with kilometers over miles, $10/6.2 = \frac{k}{15}$.

$k = 15 \times 10 \div 6.2 \approx 24.194$, which rounds to 24.2 kilometers.

Time: the pace is a second rate, 6 minutes for each kilometer, so the time is about $24.194 \times 6 \approx 145.16$ minutes, which rounds to 145 minutes (2 hours 25 minutes).

Using the rounded 24.2 kilometers gives 145.20, which rounds the same way.

Re-check strip (3 items, C-AR.3.2-R)

1. Regular **B. 11.25 ft**

The tree's shadow is $20 \div 8 = 2.5$ times as long as the person's.

The tree is 2.5 times as tall: $4.5 \times 2.5 = 11.25\,\text{ft}$.

2. Regular **B. 37.50 km**

Scale: $15 \div 4 = 3.75\,\text{km}$ for every 1 cm.

$10 \times 3.75 = 37.50\,\text{km}$.

Adding 6 cm to 15 km treats the scale as additive; $4 \times 10 \div 15$ sets up the ratio upside down.

3. Regular **17.5 gallons**

Unit rate: $5.25 \div 3 = 1.75$ gallons per hour.

$10 \times 1.75 = 17.5$.

Comparison strip (2 items, C-AR.3.2-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular $\frac{3}{4}$ **cup**

Oil for 1 dozen: $\frac{1}{3} \div 4 = \frac{1}{12}$ cup.

Oil for 9 dozen: $9 \times \frac{1}{12} = \frac{3}{4}$ cup.

2. Regular **C. 9.60 km**

Scale: $8 \div 5 = 1.6\,\text{km}$ for every 1 cm.

$6 \times 1.6 = 9.60\,\text{km}$.

Adding 1 cm to 8 km treats the scale as additive; $5 \times 6 \div 8$ sets up the ratio upside down.

Reteach plan — C-AR.3.2

MA.7.AR.3.2 · Solve problems involving proportions

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

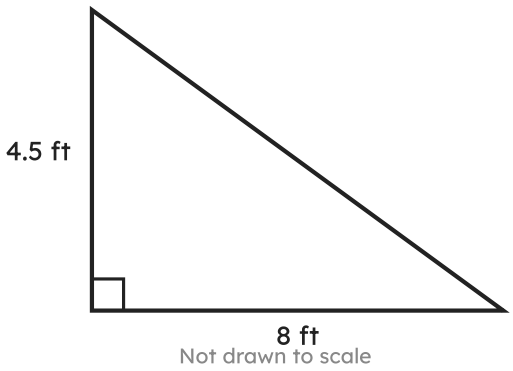
Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-AR.3.2-R

Benchmark: **MA.7.AR.3.2** · Solve problems involving proportions Student code: _____ Calculator: yes

1. A 4.5-foot person casts an 8-foot shadow. At the same time a tree casts a 20-foot shadow.

How tall is the tree, in feet?



Not drawn to scale

A. 35.56 ft

B. 11.25 ft

C. 4.44 ft

D. 16.50 ft

2. On a map, 4 cm represents 15 km.

How many kilometers does 10 cm represent?

A. 21.00 km

B. 37.50 km

C. 2.67 km

D. 150.00 km

3. A drip line uses 5.25 gallons of water every 3 hours.

How many gallons of water does the drip line use in 10 hours?

Score: ____ / 3 → copy to your progress strip

7th Grade Math · Comparison items (2) C-AR.3.2-X

Benchmark: **MA.7.AR.3.2** · Solve problems involving proportions Student code: _____ Calculator: yes

1. A recipe uses $\frac{1}{3}$ cup of oil for every 4 dozen muffins. The amounts are proportional.

How much oil is needed for 9 dozen muffins? Write your answer as a fraction or a mixed number in simplest form.

2. On a map, 5 cm represents 8 km.

How many kilometers does 6 cm represent?

A. 3.75 km

B. 48.00 km

C. 9.60 km

D. 9.00 km

Score: ____ / 2 → **copy to your progress strip**

7th Grade Math · Comparison items (2) C-AR.3.2-X

Benchmark: **MA.7.AR.3.2** · Solve problems involving proportions Student code: _____ Calculator: yes

1. A recipe uses $\frac{1}{3}$ cup of oil for every 4 dozen muffins. The amounts are proportional.

How much oil is needed for 9 dozen muffins? Write your answer as a fraction or a mixed number in simplest form.

2. On a map, 5 cm represents 8 km.

How many kilometers does 6 cm represent?

A. 3.75 km

B. 48.00 km

C. 9.60 km

D. 9.00 km

Score: ____ / 2 → **copy to your progress strip**